International Journal of Sports Physiology and Performance, (Ahead of Print)
https://doi.org/10.1123/ijspp.2023-0273
© 2024 Human Kinetics, Inc.
First Published Online: Jan. 18, 2024

Human Kinetics
ORIGINAL INVESTIGATION

Why Should Athletes Brake Fast? Influence of Eccentric Velocity on Concentric Performance During Countermovement Jumps at Different Loads

Jose L. Hernández-Davó,[1,2] Rafael Sabido,[2] Manuel Omar-García,[2] and Daniel Boullosa[3,4,5]

[1]Department of Health Sciences, Universidad Isabel I, Burgos, Spain; [2]Department of Sports Sciences, Miguel Hernandez University of Elche, Elche, Spain; [3]Faculty of Physical Activity and Sports Sciences, Universidad de León, Leon, Spain; [4]Integrated Institute of Health, Federal University of Mato Grosso do Sul, Campo Grande, Brazil; [5]College of Healthcare Sciences, James Cook University, Townsville, Australia

Purpose: The aim of the present study was to analyze the effect of different eccentric tempos on eccentric kinetics and kinematics and the subsequent concentric performance when performing countermovement jumps against different loads. ***Methods:*** After 1-repetition-maximum assessment and 2 familiarization sessions, 13 well-trained participants performed, in randomized order, 12 sets (4 tempos × 3 loads) of 4 repetitions of the loaded countermovement-jump exercise. The eccentric tempos analyzed were 5 and 2 seconds, as fast as possible, and accelerated (ie, without pause between repetitions), while the loads used were 30%, 50%, and 70% of 1-repetition maximum. Several kinetic and kinematic variables during both phases were recorded by linking a linear position transducer to the barbell. ***Results:*** The eccentric work was greater in the accelerated condition despite no changes in the eccentric depth. The peak and mean propulsive velocities were greater in the as-fast-as-possible and accelerated conditions. Correlation analysis showed that, compared with the 5-second condition, the increased concentric performance in the accelerated condition was related to the difference in eccentric work performed in the last 100 milliseconds of the eccentric phase ($r > .770$). ***Conclusions:*** Contrary to current practices, the current study highlights the need for performing the eccentric phase of loaded countermovement jumps, a common exercise performed by athletes for both training and evaluation purposes, as fast as possible. This allows not only a greater eccentric work but also improved concentric performance.

Keywords: stretch-shortening cycle, muscle power, athletic performance, velocity-based training, force–velocity profile

The development of strength and power capabilities is one of the main objectives among scientists and strength and conditioning practitioners due to its crucial role in both health and sports performance. When designing resistance training programs, there are many variables (eg, volume, intensity, and rest intervals) that can be manipulated based on the specific aim of the session and the desired neuromuscular chronic adaptations.[1,2] Within these programming variables, movement velocity has emerged as a key variable during the last decade.[3–5] Despite the growing research interest in the acute and chronic responses to different movement velocities, most research has been conducted focusing on the bar velocity only during the concentric phase. A possible explanation of this focus could be the greater reliability scores shown by concentric phase variables when the eccentric phase is somehow limited (eg, an imposed pause between eccentric and concentric actions).[6] However, eccentric–concentric coupled actions are more usual during both sporting actions and training programs and also influence the concentric force–velocity relationship.[7] Despite the clear influence of the eccentric phase on the load–velocity relationship,[6,8] there is a disparity of instructions about how to perform the eccentric phase during testing procedures. In this regard, previous research has used a continuous and controlled (2–3 s) eccentric duration,[6,9] a self-selected eccentric velocity,[10] a fast downward movement,[11] and a maximal voluntary velocity.[12,13] Although speculative, these different instructions may impact the athletes' load–velocity profiles, which may lead to errors in training prescription. Therefore, most conducted research on this topic should be carefully interpreted.

Some recent works have tried to provide insights into the impact of different eccentric movement tempos on mechanical, hormonal, and neuromuscular responses to resistance training.[8] In this regard, a recent review highlighted the need for further research on this topic, as there is limited evidence of how different eccentric tempos influence strength and hypertrophic adaptations.[14] Current evidence supports slow eccentric tempos as an optimal strategy to increase muscle time under tension.[8,14] This greater time under tension has been associated with increased acute metabolic (eg, lactate) and hormonal (eg, testosterone and growth hormone) responses after a resistance training session.[8] However, there is no clear evidence of the superiority of slow tempos in eliciting greater muscular hypertrophy.[4,14,15] Despite no differences between different tempos in total muscle hypertrophy, previous research has suggested that regional hypertrophy differences are possible.[16] Moreover, fast eccentric tempos favor greater forces at longer muscle lengths, leading to greater hypertrophic effects at distal muscle regions.[16] By increasing movement velocity, kinetic energy ($0.5\ m \times v^2$) will be significantly greater.[17] Further, if an athlete moves at a greater velocity, its momentum will be higher, thus increasing the force necessary to decelerate.[14,17] These biomechanical factors linked to fast eccentric actions lead to greater force produced at the end of the eccentric phase, which has been related to increased subsequent concentric performances.[14,17–19] On the contrary, slow eccentric velocities have been shown to negatively impact both velocity and power during the subsequent concentric actions.[20] Therefore, the increase of eccentric velocity would

Sabido https://orcid.org/0000-0003-0158-9075
Boullosa https://orcid.org/0000-0002-8477-127X
Hernández-Davó (jlhdez43@gmail.com) is corresponding author, https://orcid.org/0000-0001-5785-4697

Authenticated t_lambrinides@yahoo.com | Downloaded 01/28/24 05:13 PM UTC

provide some advantages for an enhanced acute muscular performance, which has been previously attributed to the role of stretch reflexes during body-weight stretch-shortening-cycle exercises.[21]

The eccentric force–velocity relationship significantly differs from the concentric one, not showing an inverse linear relationship as the latter.[22] However, greater eccentric velocities will not necessarily imply an increase in force production.[23,24] It has been recently suggested that either the external load, the contraction intensity, or the movement velocity during the eccentric phase plays a role in concentric force enhancement.[25,26] However, most of this evidence came from studies assessing eccentric-only actions using isokinetic dynamometers,[14] and research evaluating the influence of different tempos on eccentric kinetic variables during commonly used resistance training exercises is scarce. To the best of the authors' knowledge, only Van den Tillaar[27] has analyzed eccentric kinetic variables when performing a back squat with the 80% of 1-repetition maximum (1-RM), showing an increase in peak force when using greater eccentric velocities.

Further, it would be also interesting to assess whether eccentric kinetics and their influence in the subsequent concentric performances may be modulated by the external load used (ie, %1-RM). Previous research has shown that the eccentric rate of force development, eccentric force production, and eccentric/concentric force ratio are maximized when mobilizing moderate (eg, 50%–80% 1-RM) loads.[28,29] Furthermore, the concentric force enhancement caused by the stretch-shortening cycle has been shown to be reduced with increased loads.[30] However, all these studies were performed with a fixed eccentric velocity. Therefore, there is a lack of research assessing whether different eccentric velocities may differently influence concentric performances depending on the external loads used.

Therefore, the aim of the present study was to analyze the effect of 4 different eccentric tempos (5 s, 2 s, as fast as possible, and accelerated) on eccentric kinetics and kinematics and the subsequent concentric performance when performing countermovement jumps against 3 different loads (30%, 50%, and 70% 1-RM).

Methods

Participants

A total of 13 well-trained young men (age = 24.6 [3.7] y; body mass = 75.6 [6.6] kg; height = 1.77 [0.04] m; relative 1-RM = 1.80 [0.22]) were recruited for this study. An a priori sample size calculation using specialized software (G*Power, version 3.1) determined that a sample of 12 participants was appropriate for this study (effect size [ES] = 0.8; $\alpha = .05$; power = 0.8). To be included in the study, all participants had to report at least 3 years of experience in resistance training and a relative barbell back squat strength ≥1.5 times their body mass. Participants reporting a current or recent (<12 mo) history of joint injury (eg, hip and knee) were excluded from participation. During the study period, all participants were required to maintain their normal life habits. Caffeine intakes were not allowed during the 3 hours before the testing sessions. All participants were informed about the potential risks of the study before signing the informed consent, which was approved by the University Ethics Committee.

Design

A randomized within-participants repeated-measures design was used to investigate the influence of different eccentric tempos (5 s, 2 s, fast, and accelerated) on eccentric phase variables and subsequent concentric performances during the countermovement jump exercise against 3 different loads (30%, 50%, and 70% 1-RM). After the 1-RM assessment and 2 familiarization sessions, all participants completed a single-testing session, which consisted of 12 sets (4 tempos × 3 loads) of 4 repetitions of the squat jump exercise in randomized order. Several kinetic and kinematic variables were recorded and calculated by linking a linear position transducer to the barbell.

Methodology

1-RM Back Squat

After a standardized comprehensive warm-up which included 5 minutes of low-intensity cycling, dynamic stretching, and core exercises (frontal and side plank and glute bridge), participants started the specific squat warm-up consisting of a single set of 8, 6, and 4 repetitions with the 50%, 70%, and 90% of participants' body mass, respectively. Subsequently, participants progressively increased the load lifted (range 2.5–10 kg) until the 1-RM was obtained.[31] The maximum number of sets allowed was 5, and rest intervals between sets were of at least 3 minutes. The back squat exercise required the participants to rest the bar on their trapezius, with a ∼90° knee joint angle. Adequate depth of the repetition was ensured by placing a tape at the bottom position of the movement.

Testing Protocol

The testing session consisted of a single set of 4 repetitions in the countermovement jump exercise in each experimental condition. A simple randomization was used to determine the order of the sets. Specifically, participants performed 4 sets with each external load (30%, 50%, and 70% 1-RM). These 4 sets differed in the eccentric tempo for each repetition, being those eccentric phases of a 5-second duration, 2-second duration (controlled by a metronome), as fast as possible, and accelerated. The accelerated eccentric velocity was achieved by allowing the participant (body + load system) to rapidly fall with gravity immediately after a previous countermovement jump repetition (ie, consecutive jumps). In the other 3 conditions (5 s, 2 s, and fast velocity), participants performed a maximal intended jump after each eccentric phase, but there was a 3-second rest between each repetition to allow stabilization.

Data Acquisition

Kinematic data from each repetition were recorded by linking a linear position transducer to 1 end of the bar (T-Force system, Ergotech), which recorded the position of the bar with an analog-to-digital conversion rate of 1000 Hz and an accuracy of 0.0002 m. The linear transducer was interfaced to a personal computer by means of a 14-bit analog-to-digital data acquisition board, where a specialized software application (T-Force Dynamic Measurement System) automatically calculated the relevant kinetic parameters. For the eccentric phase, the variables analyzed were: eccentric depth, eccentric phase duration, eccentric total work (W_{ecc}Total), and eccentric work in the last 100 ms before the zero-velocity point (W_{ecc}100ms). Eccentric work variables were used to reflect changes in the movement pattern (eg, differences in the last moments of the eccentric phase). For the concentric phase, the variables analyzed were: concentric force in the first 100 ms (F_{conc}100ms), mean propulsive velocity (MPV), and peak velocity (PV). These concentric variables were selected because they reflect the influence of the eccentric phase on both the subsequent concentric performance

Authenticated t_lambrinides@yahoo.com | Downloaded 01/28/24 05:13 PM UTC

(F_{conc}100ms) and the jumping performance (PV). As the first repetition in the accelerated condition was not such a condition (no previous jump), all these variables were calculated using the mean values of the second, third, and fourth repetition at each condition. A high to very high validity and reliability in both kinematic and kinetic data recorded with this system have been previously reported.[32] Specifically, the reliability of this system to analyze specific time interval data has been previously examined in our laboratory, showing an intraclass correlation coefficient >.95, and a coefficient of variation <6.5%.[29]

Statistical Analysis

All data were analyzed using the statistical package SPSS (version 25.0). After confirming data normality by performing a Kolmogorov–Smirnov test, a 2-way multivariate analysis of variance (load: 3 levels × eccentric tempo: 4 levels) was used to analyze the effect of the load and the eccentric tempo on kinematic and kinetic variables. When appropriate, pairwise comparisons were performed using a Bonferroni post hoc test. Statistical significance was set at $P < .05$. In addition, Cohen *d* ESs were used to evaluate the magnitude of the differences between conditions, interpreted as small (0.2–0.5), moderate (0.5–0.8), and large (>0.8) effects. To analyze the relationships between eccentric work and concentric performance, Pearson product–moment correlation coefficient (*r*) was calculated and interpreted as trivial (<.30), small (.30–.49), moderate (.50–.70), and large (>.70).

Results

Kinematic data from the eccentric phase in each condition are shown in Table 1. There was a main effect of eccentric tempo ($P < .001$) on eccentric phase duration. There was no main effect of load ($P = .254$) or tempo (0.278) on eccentric depth. A significant tempo × load interaction on eccentric phase duration was found ($P = .039$). Thus, while eccentric duration remains unchanged during the 5- and 2-second conditions, during the Fast (ES range = 0.83–0.97) and Accelerated conditions (ES range = 1.41–2.27), the eccentric duration increased when increasing the external load.

Data from eccentric work in each tempo condition are shown in Table 2. For both W_{ecc}Total and W_{ecc}100ms, a main effect of tempo ($P < .001$) and tempo × load interaction ($P < .001$) was found for both variables. The main effect of the load was significant for W_{ecc}Total ($P < .001$) with a tendency observed for W_{ecc}100ms ($P = .067$). When comparing W_{ecc}Total between the different tempo conditions (within the same load), the Accelerated condition showed significantly greater values than all the other conditions, with ES ranging from 2.13 to 2.29 (at 30% 1-RM), from 2.48 to 2.57 (at 50% 1-RM), and from 1.67 to 1.83 (at 70% 1-RM). For W_{ecc}100ms, both the Fast and Accelerated tempo conditions showed greater values (with ES above 1.5) than the 2- and 5-second conditions. During the 2-second condition, W_{ecc}100ms was significantly higher (ES = 1.68–2.05) than the 5-second condition. When comparing the Accelerated versus Fast conditions, W_{ecc}100ms was greater in the Accelerated condition with the 30% 1-RM (ES = 0.73) and the 50%

Table 1 Comparison of Eccentric Kinematic Data by Eccentric Tempo and Load

	5 s	2 s	Fast	Accelerated
Eccentric duration, ms				
30% 1-RM	4960 (239)	1787 (160)[a]	544 (69)[a,b]	275 (56)[a,b,c]
50% 1-RM	4871 (266)	1839 (150)[a]	560 (75)[a,b]	325 (57)*[a,b,c]
70% 1-RM	4883 (221)	1810 (231)[a]	662 (157)*#[a,b]	408 (61)*#[a,b,c]
Eccentric depth, cm				
30% 1-RM	29.6 (4.4)	30.1 (4.9)	31.3 (4.6)	30.4 (6.9)
50% 1-RM	30.2 (4.4)	29.1 (5.3)	30.7 (4.4)	31.2 (5.1)
70% 1-RM	30.8 (5.2)	29.2 (5.1)	30.7 (5.5)	30.6 (6.5)

Abbreviation: 1-RM, 1-repetition maximum.
[a]Significantly different from 5 seconds. [b]Significantly different from 2 seconds. [c]Significantly different from Fast.
*Significantly different from 30% 1-RM. #Significantly different from 50% 1-RM.

Table 2 Comparison of Eccentric Work Variables by Tempo and Load

	5 s	2 s	Fast	Accelerated
W_{ecc}Total, W				
30% 1-RM	118.2 (27.7)	120.5 (31.0)	125.2 (29.5)	233.4 (65.4)[a,b,c]
50% 1-RM	201.6 (46.3)*	193.2 (49.2)*	202.6 (44.7)*	357.3 (75.8)*[a,b,c]
70% 1-RM	283.3 (59.3)*#	270.4 (63.8)*#	283.5 (68.3)*#	443.8 (117.6)*#[a,b,c]
W_{ecc}100 milliseconds, W				
30% 1-RM	5.3 (2.5)	11.2 (3.2)[a]	27.8 (9.7)[a,b]	36.9 (14.8)[a,b,c]
50% 1-RM	5.7 (2.6)	12.6 (5.2)[a]	31.2 (12.2)[a,b]	38.9 (11.8)[a,b,c]
70% 1-RM	4.3 (2.9)	12.8 (5.3)[a]	27.9 (11.1)[a,b]	29.9 (11.4)#[a,b]

Abbreviations: 1-RM, 1-repetition maximum; W_{ecc}100 milliseconds, eccentric work in the last 100 milliseconds before the zero-velocity point; W_{ecc}Total, eccentric total work.
[a]Significantly different from 5 seconds. [b]Significantly different from 2 seconds. [c]Significantly different from Fast.
*Significantly different from 30% 1-RM. #Significantly different from 50% 1-RM.

Authenticated t_lambrinides@yahoo.com | Downloaded 01/28/24 05:13 PM UTC

1-RM (ES = 0.64). Finally, during the Accelerated tempo condition, W_{ecc}100ms with the 70% 1-RM was significantly lower than with the 50% 1-RM (ES = 0.53).

An example of the work-time data from a representative participant is shown in Figure 1.

The comparisons of concentric force produced during the first 100 ms are shown in Figure 2. There were significant main effects for eccentric tempo ($P > .001$) and load ($P > .001$), as well as a significant tempo × load interaction ($P = .004$). Specifically, F_{conc}100ms was higher during the Fast (ES ranging from 0.75 to 0.88) and Accelerated (ES ranging from 0.90 to 1.01) conditions than in the 5-second condition. Greater F_{conc}100ms was also found under the Fast (ES = 0.22–0.59) and Accelerated (ES = 0.29–0.88) conditions compared with the 2-second condition. Compared with the 5-second tempo condition, during the 2-second tempo condition, greater values were found at 50% 1-RM (ES = 0.48) and 70% 1-RM (ES = 0.70). No differences were found between the Fast and Accelerated conditions (ES ranging from 0.05 to 0.18).

Similarly, for concentric PV data, the main effects of tempo and load ($P < .001$) and a tempo × load interaction ($P < .001$) were found (see Figure 3). Specifically, when comparing with the 5-second tempo condition, PV was higher during the Fast (ES ranging from 0.67 to 1.40) and Accelerated (ES ranging from 0.48 to 1.18) eccentric tempos. When comparing with 2-second eccentric tempo, greater PV was found for the Fast (ES = 0.67–1.24) and Accelerated (ES = 0.48–1.04) tempo conditions. Compared with the 5-second tempo condition, during the 2-second tempo condition, greater values were found at 50% 1-RM (ES = 0.43). No significant differences were found between the Fast and Accelerated conditions (ES ranging from 0.10 to 0.37).

In the same line, MPV data showed the main effects of tempo, load, and a tempo × load interaction ($P < .001$) (Table 3). When compared with the 5-second tempo condition, MPV was higher during the Fast (ES ranging from 1.50 to 1.79) and Accelerated (ES ranging from 1.51 to 1.88) eccentric tempos. When comparing with the 2-second eccentric tempo, greater MPV values were found for the Fast (ES = 0.40–1.44) and Accelerated (ES = 0.69–1.44) tempo conditions. Compared with the 5-second tempo condition, the 2-second tempo condition showed greater MPV values (ES = 0.85–1.19). No significant differences were found between the Fast and Accelerated conditions (ES ranging from 0.01 to 0.14).

To analyze how the differences in W_{ecc}100ms between the 5-second and Accelerated tempo conditions influenced the increases in concentric performance (F_{conc}100ms), a correlation

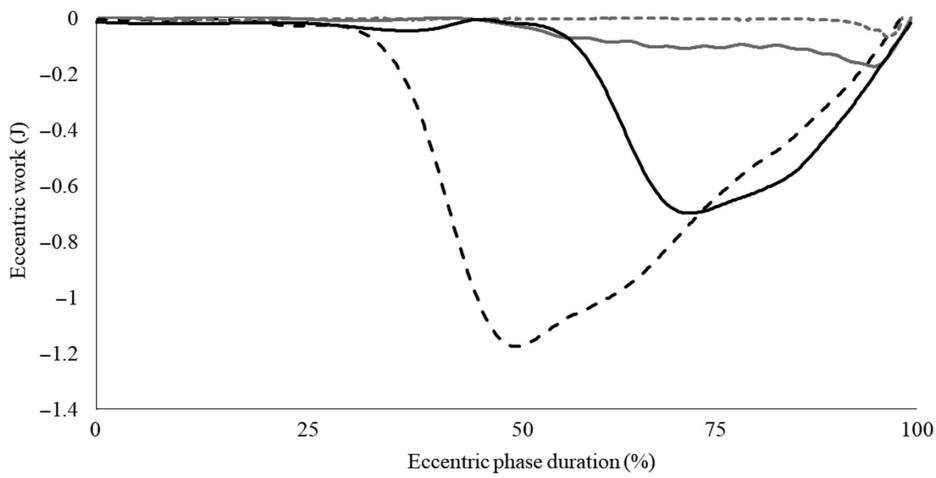

Figure 1 — Mechanical work–time data from 1 participant with a 50% 1-repetition-maximum load. Gray dotted lines = 5-second condition; gray solid line = 2-second condition; black solid line = fast condition; black dotted lines = accelerated condition.

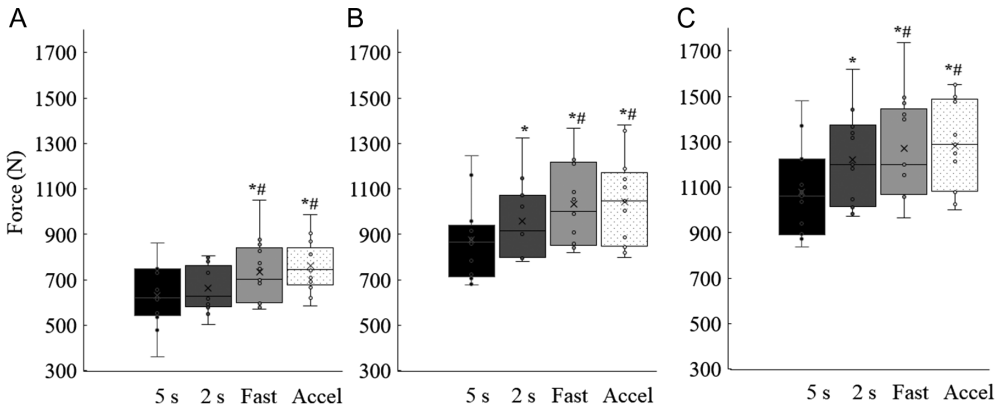

Figure 2 — Concentric force produced during the first 100 milliseconds by tempo and load. (A) 30%, (B) 50%, and (C) 70% 1-repetition maximum. Accel indicates the accelerated condition. *Significantly greater than 5-second condition. #Significantly greater than 2-second condition.

Authenticated t_lambrinides@yahoo.com | Downloaded 01/28/24 05:13 PM UTC

analysis was performed between the delta values of these variables in both conditions (Table 4). Significant large relationships were found for all loads.

Discussion

The aim of the present study was to analyze the effect of 4 different eccentric tempos (5 s, 2 s, Fast, and Accelerated) on eccentric kinetics and kinematics and the subsequent concentric performance when performing jump squats against 3 different loads (30%, 50%, and 70% 1-RM). The main findings were: (1) W_{ecc}Total is significantly increased during the Accelerated condition, (2) eccentric work in the last 100 ms of the phase is significantly increased with greater eccentric velocity conditions, (3) concentric performance is reduced when using submaximal (ie, 5 s or 2 s) tempos, and (4) increases in concentric performance are largely related to greater eccentric work in the last phase before eccentric–concentric transition.

Despite its potential relevance, research assessing the influence of different tempos on eccentric kinematic variables during coupled eccentric–concentric actions is scarce.[14] Mechanical work is considered one of the most valid methods to quantify the training volume during a resistance training session.[33] As expected, during the 5-second, 2-second, and Fast eccentric tempo conditions, W_{ecc}Total was not different. However, due to the previous jump and the increased momentum, the Accelerated condition led to greater eccentric forces, which significantly increased W_{ecc}Total (ES ranging from 1.67 to 2.57). When specifically looking at the end of the eccentric phase, W_{ecc}100ms significantly differed between conditions (see Table 2). Thus, the greater the eccentric velocity, the greater the eccentric work done at fairly similar muscle lengths. It should be highlighted that, within each tempo condition, W_{ecc}100ms did not differ between the different external loads used (30%, 50%, and 70% 1-RM). Therefore, if the aim of the training session is to maximize the mechanical work performed at long muscle lengths, eccentric velocity but not external load is critical. However, during the Accelerated tempo condition, W_{ecc}100ms was significantly lower with the 70% 1-RM compared with the 50% 1-RM. This may reflect the inability of the neuromuscular system to produce the required force to brake the body + load system in such a short time. This is in line with the study by Hernandez-Davó et al,[29] who reported that the greatest eccentric force values were found further from the eccentric–concentric transition as the external load is increased during the countermovement jump exercise. The trigger of reflex inhibition and the decrease of mechanical recoil of the muscle-tendon unit have been

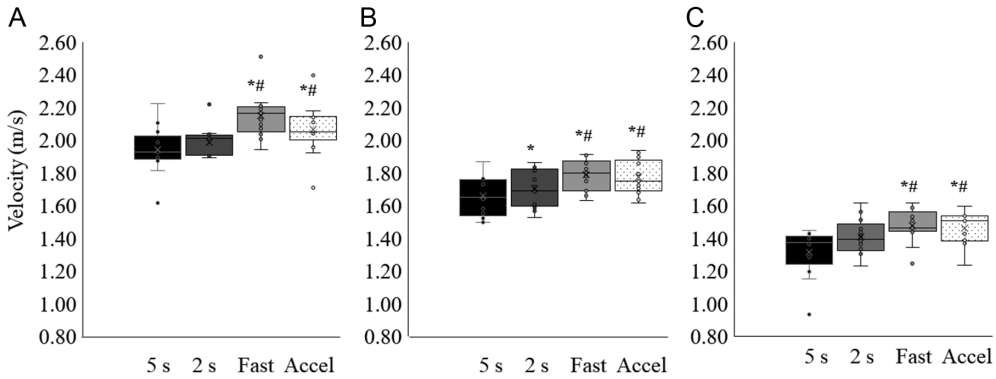

Figure 3 — Concentric peak velocity by tempo and load. (A) 30%, (B) 50%, and (C) 70% 1-repetition maximum. Accel indicates the accelerated condition. *Significantly greater than 5-second condition. #Significantly greater than 2-second condition.

Table 3 Mean Propulsive Velocity Data by Eccentric Tempo and Load

	5 s	2 s	Fast	Accelerated
30% 1-RM	1.09 + 0.10	1.16 + 0.06[a]	1.27 + 0.09[a,b]	1.27 + 0.09[a,b]
50% 1-RM	0.89 + 0.10	0.98 + 0.07[a]	1.05 + 0.07[a,b]	1.05 + 0.10[a,b]
70% 1-RM	0.66 + 0.11	0.78 + 0.09[a]	0.82 + 0.09[a,b]	0.83 + 0.05[a,b]

Abbreviation: 1-RM, 1-repetition maximum.
[a]Significantly greater than 5 seconds. [b]Significantly greater than 2 seconds.

Table 4 Correlation Analysis Between the Delta (5-Second vs Accelerated Condition) in W_{ecc}100 Milliseconds and the Delta in F_{conc}100 Milliseconds (5-Second vs Accelerated Condition) by Load

	30% 1-RM	50% 1-RM	70% 1-RM
▲ W_{ecc}100 milliseconds and ▲ F_{conc}100 milliseconds	.880*	.790*	.774*

Abbreviations: F_{conc}100 milliseconds, concentric force in the first 100 milliseconds; 1-RM, 1-repetition maximum; W_{ecc}100 milliseconds, eccentric work in the last 100 milliseconds before the zero-velocity point.
*Significant relationship ($P < .01$).

Authenticated t_lambrinides@yahoo.com | Downloaded 01/28/24 05:13 PM UTC

suggested as factors explaining this inability.[34] Nevertheless, despite the greater overall values of W_{ecc}100ms with the 50% 1-RM load, 3 of the 13 participants showed greater values with the 70% 1-RM load, which indicates that individual characteristics influence the ability to generate force in the very late portions of the eccentric phase. This is of significant practical use, as an optimal external load may differ depending not only on the specific muscle length where the muscular work wants to be maximized but also on the specific individuals' characteristics.

Jumping performance has been previously linked to the ability to produce greater muscle forces over the first part of the concentric action.[35] The present results highlight that fast eccentric velocities, as recorded in the Fast and Accelerated tempo conditions, are a key factor in producing greater concentric forces in the early moments (eg, 100 ms) of the concentric phase. Despite the differences in W_{ecc}100ms between the Fast and Accelerated tempo conditions, F_{conc}100ms was not different. This is in line with Aboodarda et al[34] who showed differences in kinetic eccentric variables when augmenting eccentric velocities through elastic bands but reported no differences in concentric performances. Of note, when compared with the 5-second tempo condition, the 2-second tempo did not lead to significantly greater F_{conc}100ms with the 30% 1-RM, but it did with the 50% and 70% 1-RM (see Figure 2). Thus, when using submaximal eccentric velocities, concentric performance is mainly affected by increases in the external load. Although speculative, it is possible that the greater effort time performed during the 5-second eccentric tempo condition may have led to some neuromuscular fatigue or inhibition, especially during the use of moderate–heavy loads, which affected the subsequent early concentric performance. When comparing PV data between the different tempos, both the Fast and Accelerated conditions led to significantly greater values than 5- and 2-second tempos with all loads. The ES for these differences showed a wide range, with the magnitudes being greater with lower loads. This would seem contrary to the results reported for F_{conc}100ms but can be explained by temporal reasons. Specifically, the difference is concentric phase times between the Accelerated and 5-second conditions changed depending on the external load used (1.0%, 3.7%, and 18.2% at 30%, 50%, and 70% 1-RM load, respectively). Thus, the higher the external load, the greater concentric movement time is required. This fact may allow the participants to compensate for the impaired concentric performance shown during the early moments of the phase, especially when using high loads. This is supported by McCarthy et al[19] who showed that the concentric phase during a stretch-shortening cycle movement is primarily potentiated during the early phases. In this regard, the difference in W_{ecc}100ms between the 5-second and Accelerated tempo conditions was calculated and related to the difference in F_{conc}100ms across these same conditions. Thus, and independently of the external load, the strength of the associations was large ($r = .774$–$.880$; Table 4) which means that early concentric performance is highly influenced by changes in W_{ecc}100ms due to the increased eccentric velocity.

The observed influence of different eccentric parameters on subsequent concentric performances confirms the need to carry out more research about eccentric actions at different velocities and their impact on acute and chronic responses to training with different exercises.[14] This is not a trivial aspect as both traditional[2] and modern resistance training approaches[3] preclude eccentric actions at maximum velocities during the execution of resistance training exercises. While traditional approaches promoted the execution of resistance training exercises at constant, well-controlled velocities during both concentric and eccentric actions (eg, 2 s during the eccentric phase and 2 s during the concentric phase),[2] the modern velocity-based training approach focuses more on performing maximal intended concentric velocities after well-controlled eccentric actions.[3] Therefore, it is unknown whether the current knowledge on the chronic effects of resistance training on strength adaptations and athletic performance may be different when considering the current findings. This is a key issue to be addressed in further studies as Cormie et al[36] previously showed that changes in the eccentric phase contribute to improved stretch-shortening cycle performance after both ballistic and heavy resistance training. Furthermore, the evaluation of both the load–velocity and the force–velocity profiles[7,37] may be importantly biased when not properly addressing the importance of eccentric velocities during different resistance training exercises. This is more evident when looking at the results obtained for the MPV, with significantly greater velocities after faster eccentric phases (see Table 3). Therefore, from the current findings, it seems evident that a precise identification and manipulation of eccentric action velocity are mandatory to better understand both the acute and chronic adaptations during resistance training exercises.

Practical Applications

To improve the concentric performance, loaded countermovement jumps should be performed with the eccentric phase as fast as possible. These findings also highlight the need for better describing the eccentric performance during loaded countermovement jump exercises and other resistance training exercises, when looking for differences in chronic adaptations between current resistance training approaches and resistance training exercises performed with maximum intended velocity during eccentric phases. For instance, most studies describing the force–velocity and load–velocity profiles of athletes have been conducted with controlled, slow eccentric tempos, thus reducing the concentric performances which result in submaximal profiles.

Conclusions

The current study clearly shows that velocity during the eccentric phase of a countermovement-jump exercise with different loads directly influenced the indices of concentric performance, with the work accumulated at the end of the eccentric phase directly related to the force developed during the early moments of the concentric phase. The maximum intended eccentric velocity resulted in the greatest concentric performance, with no differences between the Fast and Accelerated conditions despite the greater eccentric work in the accelerated condition (ie, repetition performed after a previous jump).

Acknowledgment

Daniel Boullosa was supported by grant RYC2021-031098-I funded by MCIN/AEI/10.13039/501100011033, by "European Union Next Generation EU/PRTR," and by a productivity research grant (PQ1D) from CNPq (Brazil).

References

1. Baechle TR, Earle RW, Wathen D. Resistance training. In: Earle RW, Baechle TR, eds. *Essentials of Strength Training and Conditioning*. 3rd ed. Human Kinetics; 2008:381–413.

Authenticated t_lambrinides@yahoo.com | Downloaded 01/28/24 05:13 PM UTC

2. Kraemer WJ, Ratamess NA. Fundamentals of resistance training: progression and exercise prescription. *Med Sci Sports Exerc*. 2004; 36(4):674–688. PubMed ID: 15064596 doi:10.1249/01.MSS.0000121945.36635.61
3. González-Badillo JJ, Sánchez-Medina L. Movement velocity as a measure of loading intensity in resistance training. *Int J Sports Med*. 2010;31(5):347–352. PubMed ID: 20180176 doi:10.1055/s-0030-1248333
4. Hackett DA, Davies TB, Orr R, Kuang K, Halaki M. Effect of movement velocity during resistance training on muscle-specific hypertrophy: a systematic review. *Eur J Sport Sci*. 2018;18(4):473–482. PubMed ID: 29431597 doi:10.1080/17461391.2018.1434563
5. Pareja-Blanco F, Rodríguez-Rosell D, Sánchez-Medina L, Gorostiaga EM, González-Badillo JJ. Effect of movement velocity during resistance training on neuromuscular performance. *Int J Sports Med*. 2014; 35(11):916–924. PubMed ID: 24886926 doi:10.1055/s-0033-1363985
6. Pallares JG, Sánchez-Medina L, Pérez CE, De La Cruz-Sánchez E, Mora-Rodriguez R. Imposing a pause between the eccentric and concentric phases increases the reliability of isoinertial strength assessments. *J Sports Sci*. 2014;32(12):1165–1175. PubMed ID: 24575723 doi:10.1080/02640414.2014.889844
7. Alcazar J, Csapo R, Ara I, Alegre LM. On the shape of the force-velocity relationship in skeletal muscles: the linear, the hyperbolic, and the double-hyperbolic. *Front Physiol*. 2019;10:769. doi:10.3389/fphys.2019.00769
8. Wilk M, Tufano JJ, Zajac A. The influence of movement tempo on acute neuromuscular, hormonal, and mechanical responses to resistance exercise—a mini review. *J Strength Cond Res*. 2020;34(8):2369–2383. PubMed ID: 32735429 doi:10.1519/JSC.0000000000003636
9. Gantois P, Fonseca FS, Nakamura FY, Fortes LS, Fernandez-Fernandez J, Batista GR. Analysis of velocity- and power-load relationships of the free-weight back-squat and hexagonal bar deadlift exercises. *Biol Sport*. 2023;40(1):201–208. PubMed ID: 36636177 doi:10.5114/biolsport.2023.112966
10. Pérez-Castilla A, Jukic I, Janicijevic D, Akyildiz Z, Senturk D, García-Ramos A. Load–velocity relationship variables to assess the maximal neuromuscular capacities during the back-squat exercise. *Sports Health*. 2022;14(6):885–893. PubMed ID: 35114871 doi:10.1177/19417381211064603
11. Smajla D, Spudić D, Kozinc Ž, Šarabon N. Differences in force-velocity profiles during countermovement jump and flywheel squats and associations with a different change of direction tests in elite karatekas. *Front Physiol*. 2022;13:828394. PubMed ID: 35800347 doi:10.3389/fphys.2022.828394
12. Pérez-Castilla A, García-Ramos A, Padial P, Morales-Artacho AJ, Feriche B. Load-velocity relationship in variations of the half-squat exercise: influence of execution technique. *J Strength Cond Res*. 2020;34(4):1024–1031. PubMed ID: 28885389 doi:10.1519/JSC.0000000000002072
13. Pérez-Castilla A, Jiménez-Reyes P, Haff GG, García-Ramos A. Assessment of the loaded squat jump and countermovement jump exercises with a linear velocity transducer: which velocity variable provides the highest reliability? *Sports Biomech*. 2021;20(2):247–260. PubMed ID: 30644796 doi:10.1080/14763141.2018.1540651
14. Handford MJ, Bright TE, Mundy P, Lake J, Theis N, Hughes JD. The need for eccentric speed: a narrative review of the effects of accelerated eccentric actions during resistance-based training. *Sports Med*. 2022;52(9):2061–2083. PubMed ID: 35536450 doi:10.1007/s40279-022-01686-z
15. Schoenfeld BJ, Ogborn DI, Krieger JW. Effect of repetition duration during resistance training on muscle hypertrophy: a systematic review and meta-analysis. *Sports Med*. 2015;45(4):577–585. PubMed ID: 25601394 doi:10.1007/s40279-015-0304-0
16. Pearson J, Wadhi T, Barakat C, et al. Does varying repetition tempo in a single-joint lower body exercise augment muscle size and strength in resistance-trained men? *J Strength Cond Res*. 2022;36(8):2162–2168. PubMed ID: 34351728 doi:10.1519/JSC.0000000000003953
17. Hernández-Davó JL, Sabido R, Behm DG, Blazevich AJ. Effects of resistance training using known vs unknown loads on eccentric-phase adaptations and concentric velocity. *Scand J Med Sci Sport* 2018;28: 407–417. doi:10.1111/sms.12933
18. Lafaye G, Wagner P. Eccentric rate of force development determines jumping performance. *Comput Methods Biomech Biomed Engin*. 2013;16:1–2. doi:10.1080/10255842.2013.815839
19. McCarthy JP, Wood DS, Bolding MS, Roy JL, Hunter GR. Potentiation of concentric force and acceleration only occurs early during the stretch-shortening cycle. *J Strength Cond Res* 2012;26(9):2345–2355. PubMed ID: 22692115 doi:10.1519/JSC.0b013e3182606cc5
20. Wilk M, Golas A, Krzysztofik M, Nawrocka M, Zajac A. The effects of eccentric cadence on power and velocity of the bar during the concentric phase of the bench press movement. *J Sports Sci Med*. 2019;18(2):191–197. PubMed ID: 31191087
21. Komi PV. Stretch-shortening cycle. In: Komi PV, ed. *Strength and Power in Sport*. 2nd ed. Blackwell Science; 2000:184–202.
22. Haeufle DFB, Günther M, Bayer A, Schmitt S. Hill-type muscle model with serial damping and eccentric force-velocity relation. *J Biomech*. 2014;47(6):1531–1536. PubMed ID: 24612719 doi:10.1016/j.jbiomech.2014.02.009
23. Chapman D, Newton M, Nosaka K. Eccentric torque-velocity relationship of the elbow flexors. *Isokinet Exerc Sci*. 2005;13(2):139–145. doi:10.3233/IES-2005-0192
24. Duncan PW, Chandler JM, Cavanaugh DK, Johnson KR, Buehler AG. Mode and speed specificity of eccentric and concentric exercise training. *J Orthop Sports Phys Ther*. 1989;11(2):70–75. PubMed ID: 18796927 doi:10.2519/jospt.1989.11.2.70
25. Groeber M, Stafilidis S, Seiberl W, Baca A. Contribution of stretch-induced force enhancement to increased performance in maximal voluntary and submaximal artificially activated stretch-shortening muscle action. *Front Physiol*. 2020;11:592183. PubMed ID: 33281623 doi:10.3389/fphys.2020.592183
26. Tomalka A, Weidner S, Hahn D, Seiberl W, Siebert T. Power amplification increases with contraction velocity during stretch-shortening cycles of skinned muscle fibers. *Front Physiol*. 2021;12: 644981. PubMed ID: 33868012 doi:10.3389/fphys.2021.644981
27. Van den Tillaar R. Effect of descent velocity upon muscle activation and performance in two-legged free weight back squats. *Sports*. 2019; 7(1):15. PubMed ID: 30621028 doi:10.3390/sports7010015
28. Bogdanis GC, Tsoukos A, Brown LE, Veligekas P, Spongos K, Terzis G. Muscle fiber and performance changes after fast eccentric complex training. *Med Sci Sports Exerc*. 2018;50(4):729–738. PubMed ID: 29206782 doi:10.1249/MSS.0000000000001507
29. Hernández-Davó JL, Sabido R, Blazevich AJ. High-speed stretch-shortening cycle exercises as a strategy to provide eccentric overload during resistance training. *Scand J Med Sci Sports*. 2021;31(12): 2211–2220. PubMed ID: 34536969 doi:10.1111/sms.14055
30. Loturco I, Mcguigan MR, Freitas TT, et al. Squat and countermovement jump performance across a range of loads: a comparison between Smith machine and free weight execution modes in elite sprinters. *Biol Sport*. 2022;39(4):1043–1048. PubMed ID: 36247935 doi:10.5114/biolsport.2022.112085
31. Earle RW, Baechle TR. *NSCA's Essentials of Personal Training*. Human Kinetics; 2004.
32. Garnacho-Castaño M, Lopez-Lastra S, Mate-Muñoz JL. Reliability and validity assessment of a linear position transducer. *J Sports Sci Med*. 2015;14:128–136. PubMed ID: 25729300

Authenticated t_lambrinides@yahoo.com | Downloaded 01/28/24 05:13 PM UTC

33. McBride JM, McCaulley GO, Cormie P, Nuzzo JL, Cavill MJ, Triplett NT. Comparison of methods to quantify volume during resistance exercise. *J Strength Cond Res*. 2009;23(1):106–110. PubMed ID: 19130641 doi:10.1519/JSC.0b013e31818efdfe
34. Aboodarda SJ, Page PA, Behm DG. Eccentric and concentric jumping performance during augmented jumps with elastic resistance: a meta-analysis. *Int J Sports Phys Ther*. 2015;10(6):839–849. PubMed ID: 26618063
35. Bobbert MF, Gerritsen KG, Litjens MC, Van Soest AJ. Why is countermovement jump height greater than squat jump height? *Med Sci Sports Exerc*. 1996;28(11):1402–1412. PubMed ID: 8933491 doi:10.1097/00005768-199611000-00009
36. Cormie P, McGuigan MR, Newton RU. Changes in the eccentric phase contribute to improved stretch-shorten cycle performance after training. *Med Sci Sport Exerc*. 2010;42(9):1731–1744. doi:10.1249/MSS.0b013e3181d392e8
37. Jiménez-Reyes P, Samozino P, Cuadrado-Peñafiel V, Conceição F, González-Badillo JJ, Morin JB. Effect of countermovement on power-force-velocity profile. *Eur J Appl Physiol*. 2014;114(11):2281–2288. PubMed ID: 25048073 doi:10.1007/s00421-014-2947-1

Authenticated t_lambrinides@yahoo.com | Downloaded 01/28/24 05:13 PM UTC